

ZHUOQING ZHONG

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EDUCATION

Carnegie Mellon University

M.S. in Electrical and Computer Engineering
Spring 2027 entering class

Wenzhou-Kean University

B.Sc. in Computer Science

Aug 2022 - Jun 2026

- Overall GPA: 3.93/4.00; Minor: Mathematical Sciences
- Honors (2024-2025): Zhejiang Province Government Scholarship; Dean's First-class Scholarship; Dean's Scholarship for Research and Innovation.

PUBLICATIONS

- Yifan Shen, Chuanmiao Dong, **Zhuoqing Zhong**, et al. "Position: Multimodal LLMs Should Learn from Children", CVPR Workshop on Computer Vision for Children (CV4CHL), 2026, pp. 5619-5630.
- Bowen Li*, **Zhuoqing Zhong***, Hengyu Zhang, Xubin Sun, Baha Ihnaini. "Improving Negative Rejection Ability in Language Models: A Review of Fine-Tuned LLMs, RAG, and RAFT", Journal of King Saud University Computer and Information Sciences, 38, 21, 2026.
- Ebenezer Fiifi Emire Atta Mills*, Zihui Deng*, **Zhuoqing Zhong***, Jinger Li. "Data-driven prediction of soccer outcomes using enhanced machine and deep learning techniques", Journal of Big Data, 2024.
- Baha Ihnaini, Wei Liu, **Zhuoqing Zhong** (co-first author), Zihui Deng, Jiayi Xu, Ruoling Zhao. "Evaluating Mild Cognitive Impairment Using Large Language Models: A Comparative Analysis", Submitted to Acta Pharmacologica Sinica.

* Equal contribution / shared first authorship.

PROFESSIONAL EXPERIENCE

Institute of Software, Chinese Academy of Sciences

Research Intern, Artificial Intelligence Algorithm Integrated Innovation Center

Aug 2024 - Sep 2024

- Implemented a data pipeline and token-level attribution visualizations using Layer-wise Relevance Propagation (LRP) and DeepLIFT for language model interpretability.
- Compared attribution methods on text examples, including medical text, and visualized token relevance to examine model behavior.

RESEARCH EXPERIENCE

RepoFusion-in-Decoder

Efficient cross-file code completion via lightweight encoder fusion

Apr 2025 - Aug 2025

- Implemented a lightweight encoder and cross-attention modules to integrate retrieved repository context into a frozen decoder-only code language model.
- Built a BM25 retrieval and sliding-window chunking pipeline, with cached encoder states to reuse cross-file context.
- Evaluated completion quality and prefill latency on Long Code Arena and RepoEval-Updated; conducted ablations on cross-attention depth to study the efficiency-quality trade-off.

Large Language Models for Mild Cognitive Impairment Screening

Co-first author | Advisor: Dr. Baha Ihnaini

Feb 2024 - Jul 2025

- Built a speech-to-text screening pipeline with audio normalization and transcript correction; curated a Chinese-language data subset and implemented balanced 5-fold cross-validation.
- Fine-tuned Llama-3, Gemma, and Qwen models using supervised fine-tuning and LoRA; compared unquantized, 8-bit, and 4-bit configurations.
- Compared zero-shot, few-shot, and instruction prompting, and evaluated model configurations using accuracy, F1, precision, and recall.

Position: Multimodal LLMs Should Learn from Children

Collaborative survey and conceptual framework

Nov 2025 - Mar 2026

- Surveyed developmental learning for multimodal models, focusing on grounding, supervision inference, memory, and continual adaptation.
- Contributed to a framework spanning task formation, supervision inference, and continuous growth; analyzed limitations in sequential multimodal data and evaluation benchmarks.

Language Model Reliability and Rejection Behavior

Co-first author | Advisor: Dr. Baha Ihnaini

Nov 2024 - May 2025

- Conducted a literature review across five scholarly databases, applying inclusion criteria, citation tracing, and structured extraction of models, datasets, and metrics.
- Compared fine-tuning, RAG, and RAFT for rejection behavior, hallucination mitigation, computational cost, and domain adaptability; co-developed a four-dimensional review framework.

Reinforcement Learning for UAV Planning

Research project

Sep 2024 - Oct 2025

- Implemented an enhanced Noisy DQN and an evaluation pipeline for a communication-constrained 15 x 15 grid environment, with comparisons against DQN, NoisyNet, and Double DQN.

CT Image Reconstruction

Research project

Feb 2025 - Jun 2025

- Implemented a U-Net with kernel basis attention and multi-axis feature fusion; evaluated low-dose and sparse-view reconstruction using MSE, SSIM, and a CNN observer task.

INTERESTS & SKILLS

- Programming and environment: Python, Java, Linux.
- Research methods: LLM fine-tuning, LoRA, quantization, retrieval, cross-validation, ablation studies, token attribution.
- Research Interests: Multimodal Large Language Models; Computer Vision; Natural Language Processing; Reinforcement Learning; Reliable Large Language Models.